

Assignment 1: Star Cluster Structure and Modeling

This assignment guides you through the analysis, modeling, and validation of a star cluster using the AMUSE framework. You will explore real cluster data, fit a King model, validate your fitting routine, and assess the dynamical state and expected lifetime of the cluster.

1. Install AMUSE on your laptop

2. Explore the provided star cluster data

Download the file `star_cluster_001.amuse` from the course website.

Write a Python/AMUSE script that:

- Reads the particle set.
- Computes and plot the radial number density profile $n(r)$
- Annotate the figure with cluster parameters, including: core radius, half-mass radius, and tidal radius. Include clear axis labels, units, and short captions explaining the physical meaning of the figure.
- Compute the cluster's virial ratio, and estimate the survival time of the cluster on a Solar-circle orbit in the Galactic tidal field.

Deliverable: The annotated plot with caption.

3. Fit a King model to the cluster

Write a Python/AMUSE routine that:

- Reads `star_cluster_001.amuse`.
- Fits a King model to extract the concentration parameter w_0 , core radius r_c , and tidal radius r_t .
- Produces a plot showing the data and the fitted King model.

Comment on the quality of the fit and the physical plausibility of the parameters.

Deliverable: An annotated figure with caption of the data and overplotted the King-model fit.

4. Validate your King fitting routine

Write a small AMUSE script that:

- Generates a synthetic King-model cluster with chosen w_0 , r_c , for N particles.
- Saves the synthetic cluster as an AMUSE particle set.
- Applies your fitting routine to recover w_0 , r_c , and r_t .

Compare the recovered parameters to the input values and discuss the accuracy and limitations of your method.

Deliverable: In class you may be asked to comment on your figure and fit.

5. What to hand in

Submit on **Monday 5 October** * PDF document description of your findings containing * Two figures (1 page). * Explanation, methods, and interpretation (1 page).